

STEFAN ANTONIO CIRISANU

Lead Blockchain Engineer & Systems Architect

- A rigorously analytical and mathematically-driven Systems Architect with over 8 years of software engineering experience and 4+ years of deep specialization in cryptographic infrastructure, Zero-Knowledge Proofs, and scalable Web3 backends. Proven ability to architect and single-handedly execute massive-scale, complex distributed systems from the ground up—bridging traditional fiat payment rails with decentralized EVM environments and Noir-based ZK circuits.

-Highly experienced in taking uncompromising ownership of product lifecycles, enforcing zero-trust API boundaries, and delivering zero-latency data pipelines. Operating at the highest echelon of full-stack engineering, I bring an uncompromising approach to security, system determinism, and infrastructure scalability, aligning perfectly with the technical leadership requirements at Metatheory.

Humanity Ledger | European Union

Chief Executive Officer & Lead Systems Architect | Jan 2026 – Present Directing the strategic and technical deployment of a next-generation decentralized ecosystem. Architected an enterprise-grade hybrid SaaS-to-Blockchain platform (HumanIDFi / Whale Network), engineering the underlying architecture with absolute determinism and zero-trust security.

Designed and deployed an asynchronous synchronization engine bridging Stripe webhooks with PostgreSQL, translating complex fiat billing logic into immutable on-chain feature gating and API quotas. Implemented mathematically rigorous API boundaries, eliminating client-side trust vectors through deep backend verification of SIWE (Sign-In with Ethereum) sessions and dynamic tier enforcement. Orchestrated the full-stack pipeline, ensuring zero-latency communication between EVM smart contracts, backend databases, and custom React dashboards rendering real-time cryptographic metrics.

Aztec Network | Remote

Independent ZK Infrastructure Developer | Feb 2026 – Present Leading the technical implementation of "Whale Network," a privacy-centric platform demonstrating the commercial viability of zero-knowledge rollups and the Noir programming language.

Engineered a modular proving abstraction layer capable of processing WASM-based client-side proofs alongside GPU-delegated server-side proofs for high-throughput institutional clients. Designed a custom Paymaster architecture, executing ECDSA signature validation on the backend to dynamically subsidize gas fees based on strict, database-enforced subscription tier quotas. Implemented sophisticated Graph Database architectures to efficiently manage complex decentralized data flows and privacy-preserving identity relationships.

Tools for Humanity Independent Cryptographic World ID Builder

Jan 2022 – Jul 2025 Contributed to the core infrastructure of global decentralized identity systems, focusing on Sybil-resistance and cryptographic privacy over a three-year tenure.

Engineered integration layers for the World ID protocol, establishing privacy-first identity verification frameworks across decentralized applications. Audited, optimized, and deployed highly secure Solidity smart contracts designed to interface seamlessly with identity oracles and zero-knowledge state verifiers. Operated as an autonomous infrastructure contractor, delivering critical technical components under uncompromising security standards and aggressive deadlines.

Wolfpack Digital | Remote

Software Engineer | Jan 2020 – Jan 2022

Spearheaded the architectural migration of monolithic backend systems into high-performance, highly concurrent Node.js/TypeScript microservices. Optimized relational database queries within high-traffic PostgreSQL environments, significantly reducing payload latency and system overhead. Engineered responsive and robust frontend architectures using React, ensuring tight synchronization with asynchronous backend states.



Skills

Distributed Ledger Technologies: Solidity, EVM Architecture, Ethereum, Aztec Network, Zero-Knowledge Proofs (zk-SNARKs), Noir Programming, Account Abstraction (ERC-4337), Custom Paymaster Design.

Backend & API Infrastructure: Node.js, Next.js 15 (App Router), TypeScript, PostgreSQL, Prisma ORM, Graph Database Schema Design, High-Frequency WebSocket Streams, FIX Protocol.

Cryptographic Security: Zero-Trust API Enforcement, Anti-Replay Mechanisms (Nonce/Timestamp Validation), ECDSA Signature Verification, Deterministic Payload Validation, Threat Modeling.

Frontend Architecture: React, Context API, TailwindCSS, Zero-Latency UI Synchronization, Complex SVG Data Visualization, SSR/SSG.

Language Proficiencies: TypeScript, JavaScript, Solidity, Noir, Rust (Working Proficiency), Python.

Education

Oracle Certified Professional

Java SE 21 Developer Certification Oracle Certified

2022 – 2023

Polytechnic University of Bucharest (UPB) Certified Degree

Cibersecurity

sept. 2018 – jun. 2022

Languages

English



Romanian



Spanish

